

SMART INDIA HACKATHON 2026

Problem Statement ID: SIH 26082

Theme: Clean & Green Technology | Category: Software

PROJECT REPORT

Air Pollution–Weather Coupled Forecasting System

Delhi NCR Focus

A physics-informed, explainable 72-hour air-quality forecasting and decision-support platform

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Team ID: 145733

Report status. This report describes the proposed architecture and the current prototype. Model accuracy, validation results and performance figures have not been experimentally established and are therefore not reported.

Abstract

Delhi NCR experiences recurring air-pollution episodes, yet most air-quality communication describes present conditions only. This report presents a physics-informed, explainable 72-hour air-quality forecasting and decision-support platform for Delhi NCR. The system couples air-quality observations from CPCB/public sources, meteorological data from Open-Meteo, atmospheric information from Copernicus ERA5 and satellite fire hotspots from NASA FIRMS.

An atmospheric physics engine derives indicators such as PBLH, inversion and stability, while a fire transport engine evaluates the potential influence of upwind fires using distance, wind alignment, timing and atmospheric conditions. A coupling and feature-engineering stage produces physics-informed features for an XGBoost model that forecasts PM2.5, PM10 and O3 up to 72 hours ahead. Forecasts are converted to the Indian AQI and shown on a React dashboard with an explanation layer (SHAP is the intended approach) and a confidence indication.

The current prototype provides a modular FastAPI backend and a React frontend using local demo data. Its displayed forecast is generated from a deterministic prototype timeline, its explanation is SHAP-style rather than genuine SHAP output, and its confidence indicator is not validated. Model performance has not yet been experimentally validated; the report therefore describes the design, prototype status and validation plan without claiming results.

Keywords: air-quality forecasting, Delhi NCR, PBLH, atmospheric inversion, fire transport, XGBoost, Indian AQI, explainable AI.

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1. Introduction

Air pollution in Delhi NCR is a recurring public-health and planning challenge. Existing air-quality communication mainly answers the question “what is the air quality now?”. Residents, institutions and decision makers, however, need to know what is likely to happen next, why it may change and how much confidence can be placed in that outlook.

This project proposes and prototypes a physics-informed, explainable 72-hour air-quality forecasting and decision-support platform focused on Delhi NCR. It couples air-quality observations, meteorology, atmospheric boundary-layer conditions and upwind satellite fire activity, derives physically meaningful features from them, and uses an XGBoost model to forecast PM2.5, PM10 and O3 up to 72 hours ahead. The forecasts are converted into the Indian AQI and presented on an interactive dashboard together with an explanation and a confidence indication.

Core question. “Given today’s pollution, weather, atmospheric conditions and upwind fire/stubble-burning activity, what will Delhi NCR’s air quality look like over the next 72 hours, why will it change, and how confident are we?”

Value proposition. From measuring today’s AQI to understanding and forecasting tomorrow’s air.

Conceptual chain. Observe → understand atmosphere → calculate pollutant transport → couple physical and environmental factors → XGBoost forecast → AQI, explanation and confidence → dashboard.

Status convention used in this report. Three labels are used consistently: CURRENT PROTOTYPE (what exists in the code today), PROPOSED / FINAL ARCHITECTURE (the intended design) and FUTURE VALIDATION / IMPROVEMENT (work still to be done). No accuracy, error or performance values are reported because none have been experimentally validated yet.

2. Problem Statement

Smart India Hackathon 2026 — Problem Statement ID: SIH 26082

Title: Air Pollution–Weather Coupled Forecasting System (Delhi NCR Focus)

Theme: Clean & Green Technology. **Category:** Software. **Team ID:** 145733. **Team Name:** ElevateX_1.

The task is to build a forecasting system that couples air-pollution and weather information for Delhi NCR so that future air quality can be anticipated rather than only observed.

3. Problem Analysis

The project addresses five inter-related problems.

- 1. Delayed forecasting.** Current AQI monitoring mainly communicates what is happening now, rather than a clear 72-hour view of how pollution may evolve.
- 2. Weather interaction.** Wind, Planetary Boundary Layer Height (PBLH), atmospheric inversion and atmospheric stability influence how pollution accumulates and disperses.
- 3. Pollution transport.** Upwind fire and stubble-burning activity can potentially influence Delhi NCR. Simply counting fires does not describe their potential transport influence.
- 4. Fragmented data.** Air-quality, weather, atmospheric and satellite fire information come from different sources and must be coupled for unified forecasting.
- 5. Limited explainability.** A useful system should communicate not only what pollution may look like, but also the important contributing factors and the forecast confidence.

4. Objectives

- Forecast PM2.5, PM10 and O3 for Delhi NCR up to 72 hours ahead.
- Convert predicted concentrations into pollutant-level and overall Indian AQI with category and trend.
- Integrate air-quality, meteorological, atmospheric (PBLH, inversion, stability) and satellite fire data.
- Evaluate potential upwind fire transport influence using distance, wind, timing and atmospheric conditions rather than raw fire counts.
- Generate physics-informed coupled features so that XGBoost receives meaningful environmental interactions.
- Provide an explanation layer answering “Why is the model predicting higher or lower pollution?”.
- Communicate forecast confidence as part of decision support.
- Deliver everything through an interactive Delhi NCR dashboard, in a modular design that can later extend to other Indian cities.

5. Proposed Solution

The proposed solution combines air-quality observations, meteorological information, atmospheric conditions and satellite fire activity with physics-informed feature engineering, a fire transport analysis, XGBoost forecasting, Indian AQI calculation, explainability, confidence information and interactive visualization. The system produces a 72-hour forecast for PM2.5, PM10, O3 and Indian AQI. It is focused initially on Delhi NCR but its modular architecture can be extended to other Indian cities.

Table 1: Problem versus proposed solution

Problem	Proposed solution component
Delayed forecasting	72-hour forecasting of PM2.5, PM10, O3 and Indian AQI with XGBoost
Weather interaction	Atmospheric physics engine: PBLH, inversion, stability, ventilation and mixing features
Pollution transport	Fire transport engine producing a Potential Fire Transport Influence indicator
Fragmented data	Ingestion, cleaning, validation and temporal/spatial alignment of CPCB, Open-Meteo, ERA5 and NASA FIRMS data
Limited explainability	Explanation layer (SHAP-based as the intended approach) and a confidence indicator

6. Existing Situation and Proposed Approach

Public air-quality services generally report present conditions. Meteorological data, boundary-layer information and satellite fire detections exist in separate portals and formats, and a user must combine them mentally to judge how air quality may evolve. Fire information is often communicated as a count of hotspots, which does not capture whether smoke could actually be carried toward Delhi NCR.

The proposed approach moves from isolated observation to coupled analysis: it aligns the sources, derives atmospheric and transport features, forecasts with a tabular machine-learning model, and explains the result. The approach is deliberately lightweight and suitable for a student-level prototype.

Positioning relative to WRF-Chem

This project is not WRF-Chem. WRF-Chem is a sophisticated numerical weather and atmospheric-chemistry modelling framework (Grell et al., 2005). The present work is a lightweight, physics-informed machine-learning architecture inspired by the broader idea of coupling atmospheric conditions with pollutant behaviour: observed air quality, weather, atmospheric-physics-derived features and fire transport features feed XGBoost. It does not attempt to reproduce a full numerical atmospheric-chemistry simulation.

7. System Overview

The platform is organised as a pipeline of six functional stages:

- Data layer — CPCB/public air-quality data, Open-Meteo weather, Copernicus ERA5 atmospheric data and NASA FIRMS fire data.
- Preparation layer — ingestion, cleaning, validation and temporal/spatial alignment.
- Physics layer — the atmospheric physics engine and the fire transport engine.
- Feature layer — the coupling engine and feature engineering.
- Model layer — XGBoost forecasting, Indian AQI conversion, explanation and confidence.
- Presentation layer — the React dashboard served by a FastAPI backend.

The MVP does not require a proprietary sensor network; it relies on public data sources and, in the current prototype, local demo data.

8. Data Sources

The project uses the following data sources only.

Table 2: Data sources and purpose

Source	Information	Purpose
CPCB / public air-quality data	PM2.5, PM10, O3, AQI-related observations, station observations, historical pollutant information	Pollution history, targets and station-level observations
Open-Meteo	Temperature, relative humidity, wind speed and direction, pressure, precipitation, boundary-layer-related information where available	Meteorological inputs and wind fields
NASA FIRMS	Fire/hotspot locations, timing, intensity and confidence where available	Satellite fire activity for transport analysis
Copernicus ERA5	Vertical temperature and wind information, atmospheric conditions	Inversion, stability and atmospheric analysis
OpenStreetMap	Geographic map data	Map visualization on the dashboard

Fire data are used to derive a transport-based influence indicator. The system does not simply count fires and claim that they directly caused a particular pollutant concentration.

9. System Architecture

Figure 1 shows the overall architecture. Four heterogeneous sources are ingested, cleaned and aligned. The aligned data pass through the atmospheric physics engine and the fire transport engine; the coupling and feature-engineering stage combines everything into a feature set for XGBoost. The forecast pollutant concentrations are converted to Indian AQI and delivered with an explanation and a confidence indication to the Delhi NCR dashboard.

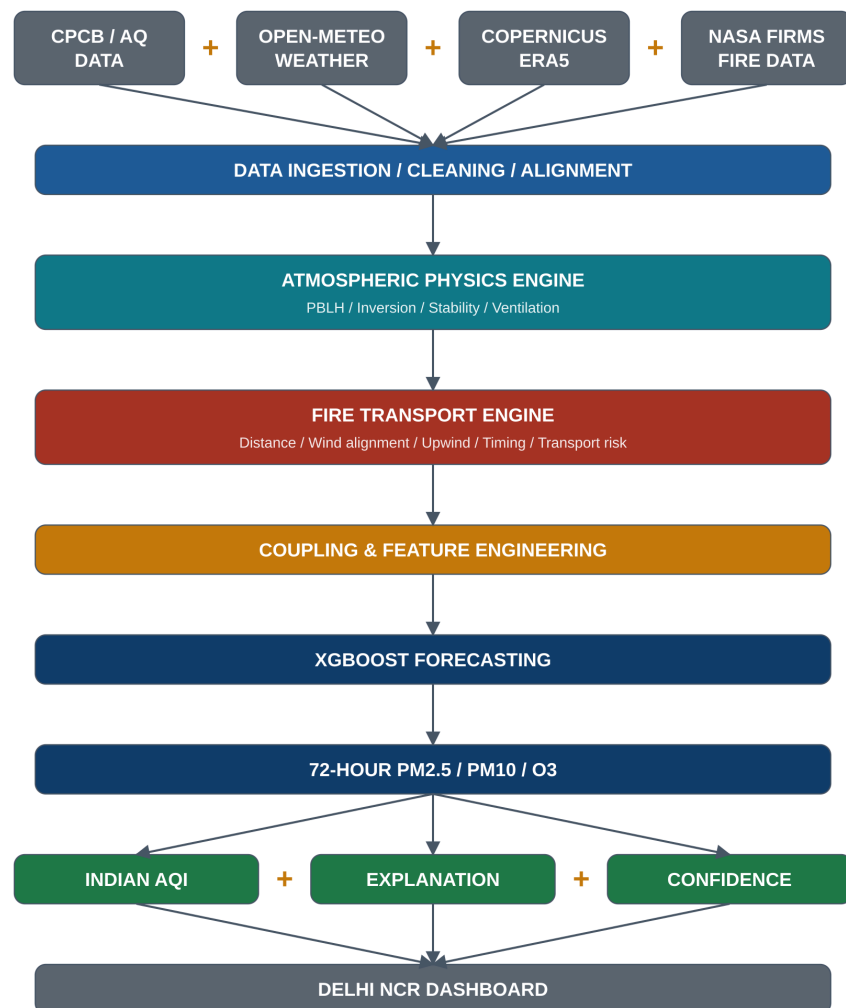


Figure 1: Overall system architecture

Architectural status. The pipeline above is the intended architecture. In the current prototype the data stage uses local demo data, and the displayed forecast is not yet produced directly by model prediction (see Chapter 29).

10. Detailed Methodology

The methodology follows the chain observe → preprocess → understand atmosphere → analyse fire transport → couple features → forecast → AQI assessment → explain → confidence → dashboard. Figure 2 shows the data flow and Figure 3 the process flow.

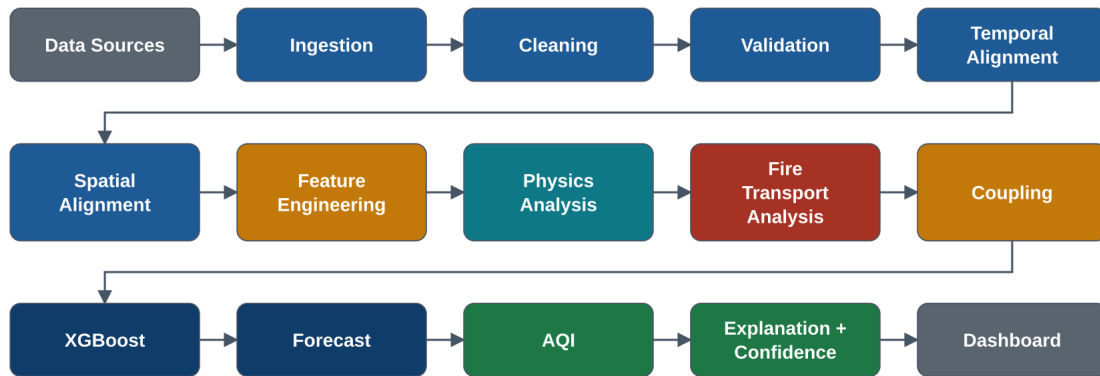


Figure 2: Data flow diagram

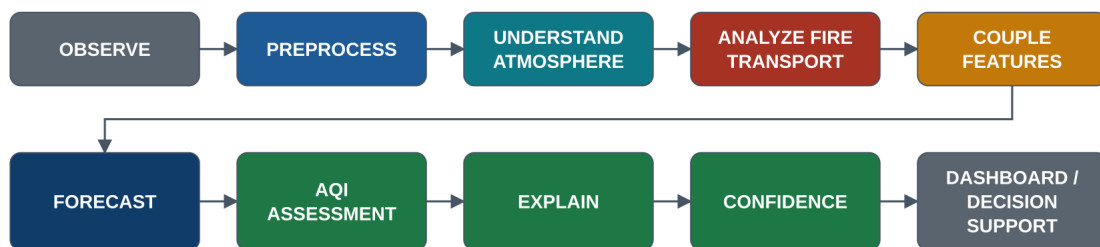


Figure 3: Methodology / process flow

Each stage is implemented as a modular component so that it can be tested and improved independently. The atmospheric analysis and fire transport analysis produce physically motivated features; the coupling stage combines them with pollution, weather and temporal information; XGBoost learns the relationship between these features and future pollutant concentrations.

11. Atmospheric Physics Analysis

A major component is the physics-informed atmospheric analysis layer. It supplies XGBoost with indicators of how easily surface emissions can mix and disperse.

Planetary Boundary Layer Height (PBLH)

PBLH represents the depth of the lower atmosphere in which surface emissions can mix. A low PBLH implies a smaller mixing volume and greater potential for pollutant accumulation; a high PBLH implies greater vertical mixing and better potential dispersion (Stull, 1988).

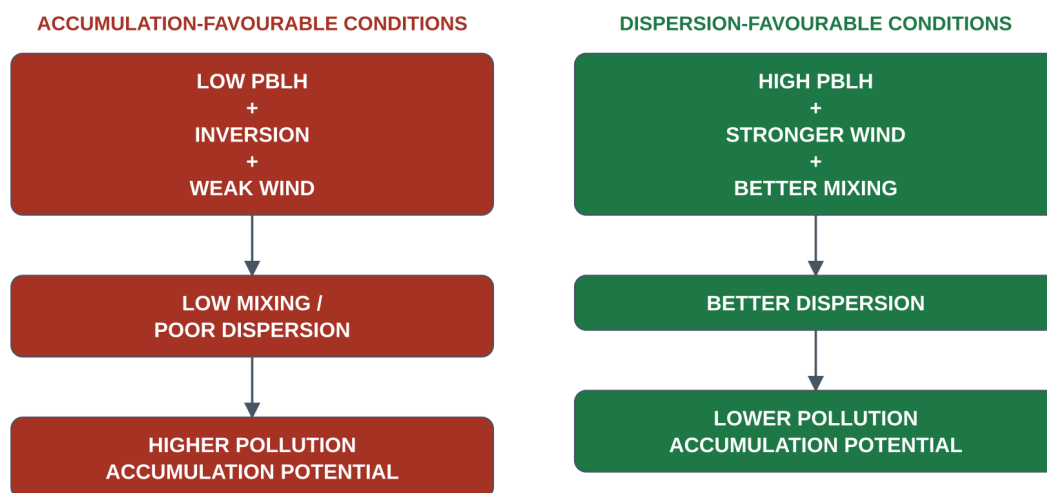
Atmospheric inversion

An inversion occurs when temperature increases with altitude over a layer, suppressing vertical mixing. The system considers inversion presence, strength and persistence together

with atmospheric stability. A combination of low PBLH, an atmospheric inversion and weak wind can indicate conditions favourable for pollution trapping.

Table 3: Atmospheric indicators

Indicator	Meaning	Pollution relevance
PBLH	Depth of the layer in which surface emissions mix	Low value: smaller mixing volume, higher accumulation potential; high value: better dispersion potential
Inversion (presence, strength, persistence)	Temperature increasing with altitude over a layer	Suppresses vertical mixing and can favour pollutant trapping
Atmospheric stability	Tendency of the atmosphere to resist vertical motion	Stable conditions reduce vertical dispersion
Wind speed	Horizontal air movement	Weak wind gives less horizontal ventilation
Ventilation potential	Derived indicator of dispersion capacity	Engineered feature for the model
Mixing potential	Derived indicator of vertical mixing	Engineered feature for the model



Physical indicators that influence pollution behaviour - not guaranteed causal predictions.

Figure 4: Atmospheric accumulation and dispersion conditions

These quantities are treated as physical indicators or features that influence pollution behaviour. They are not guaranteed causal predictions.

12. Fire Transport Analysis

NASA FIRMS provides satellite fire hotspots. Rather than counting hotspots, the fire transport engine evaluates whether fire activity could plausibly influence Delhi NCR, using the factors in Table 4. Wind vector components are derived from wind speed and direction:

$$u = \text{wind speed} \times \cos(\text{direction})$$

$$v = \text{wind speed} \times \sin(\text{direction})$$

The wind direction convention (the direction the wind blows from, as reported by meteorological sources) must be handled consistently when these components are used.

Table 4: Fire transport factors

Factor	Role
Fire location	Position of each satellite hotspot
Distance from Delhi NCR	Closer hotspots may have shorter potential transport time
Wind direction and speed	Determine the direction and pace of potential transport (via u, v components)
Upwind / downwind relationship	Indicates whether the hotspot lies on the side from which wind arrives
Wind alignment	Degree to which the wind is directed from the fire toward Delhi NCR
Timing	Relates fire detection time to the forecast window (potential impact window)
Atmospheric stability and PBLH	Modulate how smoke may mix or remain concentrated
Transport conditions	Combined context of the above factors

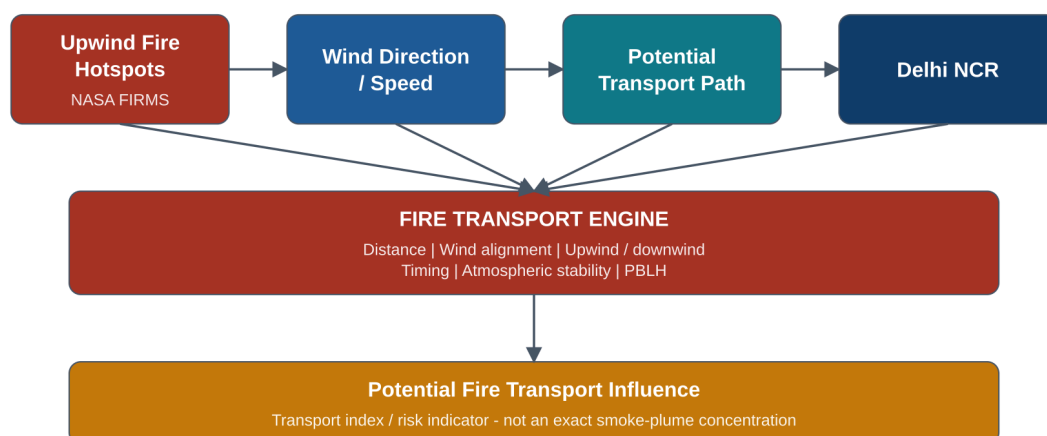


Figure 5: Fire transport concept

The engine outputs a transport or influence feature, described as “Potential Fire Transport Influence” or “Fire Transport Risk”. It is not an exact smoke-plume concentration model, and the system does not claim the pollutant concentration caused by any individual fire. The way individual factors are combined into the index has not yet been validated (see Chapter 30).

13. Data Preprocessing and Alignment

Because the sources differ in structure, resolution and update frequency, a preparation stage precedes any modelling:

- Ingestion of air-quality, weather, atmospheric and fire data.
- Cleaning and data validation to detect inconsistent or invalid records.

- Missing-data imputation where observations are absent.
- Temporal alignment so that all inputs refer to common time steps.
- Spatial alignment so that station, grid and hotspot information refer to Delhi NCR consistently.
- Use of fallback sources where a data source is temporarily unavailable.

In the current prototype these steps operate on local demo files (demo_data/air_quality.csv, weather.csv, atmosphere.csv, fires.csv) rather than live integrated feeds.

14. Feature Engineering

Feature engineering converts aligned data into the model input matrix (Figure 6). Table 5 lists the feature categories.

Table 5: Feature categories

Category	Features	Purpose
Pollution	PM2.5 lags (1h, 3h, 6h, 12h, 24h); PM10 and O3 lags; rolling averages; rolling standard deviation; rate of change; pollution persistence	Capture temporal behaviour and momentum of pollution
Weather	Temperature, relative humidity, wind speed and direction, pressure, precipitation, dew point, wind vector components, temporal/seasonal features	Describe meteorological drivers
Atmospheric	PBLH, inversion indicator, inversion strength, atmospheric stability, ventilation potential, mixing potential	Describe mixing and dispersion capacity
Fire	Hotspot count and density, distance, upwind hotspot count, fire intensity where available, timing, wind alignment, transport risk, potential impact window	Describe potential upwind fire influence
Coupling	Pollution accumulation potential, dispersion potential, inversion + wind interaction, PBLH + pollution interaction, fire + wind transport interaction, pollution persistence, atmospheric transport indicators	Encode interactions between pollution, weather and atmosphere

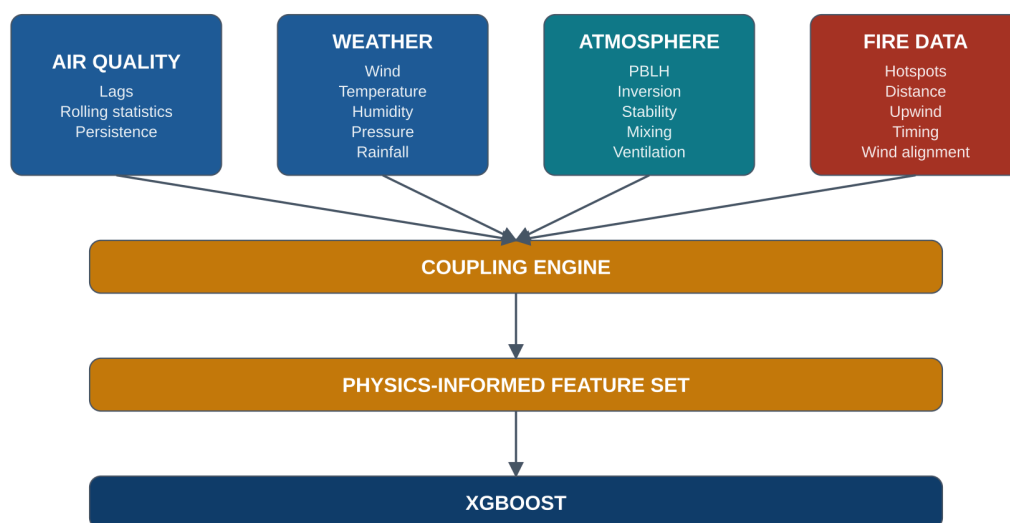


Figure 6: Physics-informed feature architecture

15. Coupling Engine

The coupling engine combines air pollution, weather, atmospheric conditions, fire activity, fire transport and temporal information. It generates features such as pollution accumulation potential, dispersion potential, ventilation potential, inversion severity, mixing potential, fire transport risk, upwind fire influence, pollution persistence and pollution–weather interaction.

Its purpose is to give XGBoost meaningful environmental interactions rather than only raw, independent variables. For example, a low PBLH matters more for accumulation when wind is also weak, and fire activity matters more when the wind is aligned toward Delhi NCR. Encoding such interactions explicitly makes the physical reasoning available to the model.

16. XGBoost Forecasting

XGBoost, a scalable gradient-boosted decision-tree method (Chen & Guestrin, 2016), is the main forecasting model. It is chosen because it suits structured/tabular environmental data, handles nonlinear relationships, combines heterogeneous features, is computationally efficient for a student-level prototype, and supports feature-importance and explainability workflows.

Model inputs are historical pollution, weather, PBLH, inversion features, atmospheric stability, fire features, fire transport features, temporal features and the physics-informed coupling features. Forecast targets are PM2.5, PM10 and O3 with a horizon of up to 72 hours; candidate evaluation horizons are 1h, 6h, 12h, 24h, 48h and 72h. No other model family (for example deep-learning models) is part of this project.

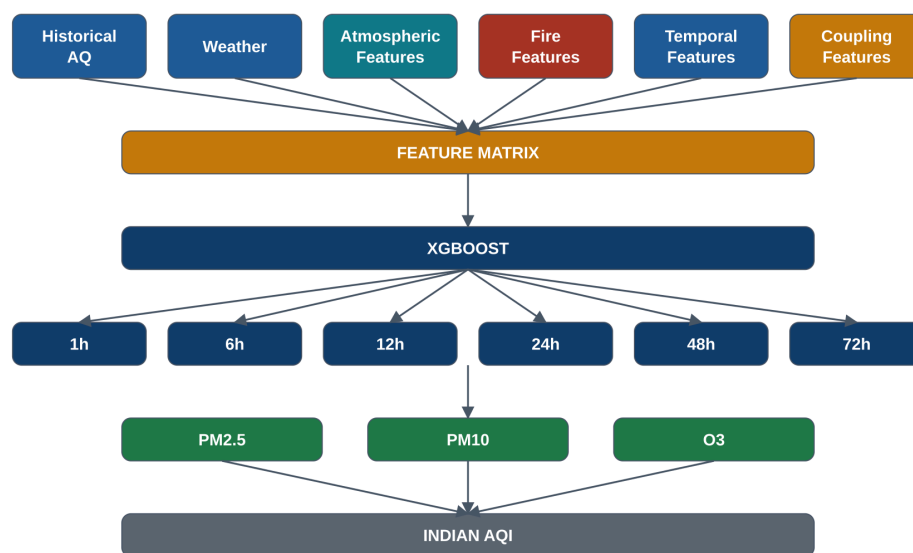


Figure 7: XGBoost forecasting pipeline

The exact multi-horizon strategy (for example one model per horizon or a single model with a horizon input) is a design decision to be finalised during implementation. No accuracy or performance figures are claimed.

17. Indian AQI Calculation

Predicted pollutant concentrations are converted into Indian AQI information using the applicable Indian AQI methodology of the Central Pollution Control Board, launched in 2014. The Indian AQI uses six categories: Good, Satisfactory, Moderately Polluted, Poor, Very Poor and Severe. The system derives pollutant-level AQI from predicted PM2.5, PM10 and O3 and an overall AQI representation.

In the standard formulation, a sub-index is obtained for each pollutant by linear interpolation between the breakpoints published by CPCB for that pollutant, and the overall AQI is determined by the worst-performing pollutant. The breakpoint tables and averaging periods are taken from the official CPCB documentation and are not reproduced or altered here.

$$I_p = ((I_{hi} - I_{lo}) / (BP_{hi} - BP_{lo})) \times (C_p - BP_{lo}) + I_{lo}$$

The dashboard communicates predicted concentrations, AQI, AQI category and forecast trend. No real AQI values are reported in this document; any values shown by the current prototype are demo values.

18. Explainable AI

The explanation layer answers “Why is the model predicting higher or lower pollution?”. Potential contributing factors include low PBLH, atmospheric inversion, weak wind, previous PM2.5 levels, upwind fire activity, higher PBLH, higher wind and rainfall/precipitation.

SHAP (Lundberg & Lee, 2017) is the intended final explainability approach. SHAP feature contributions describe model behaviour and association; they are not proof of physical causation.

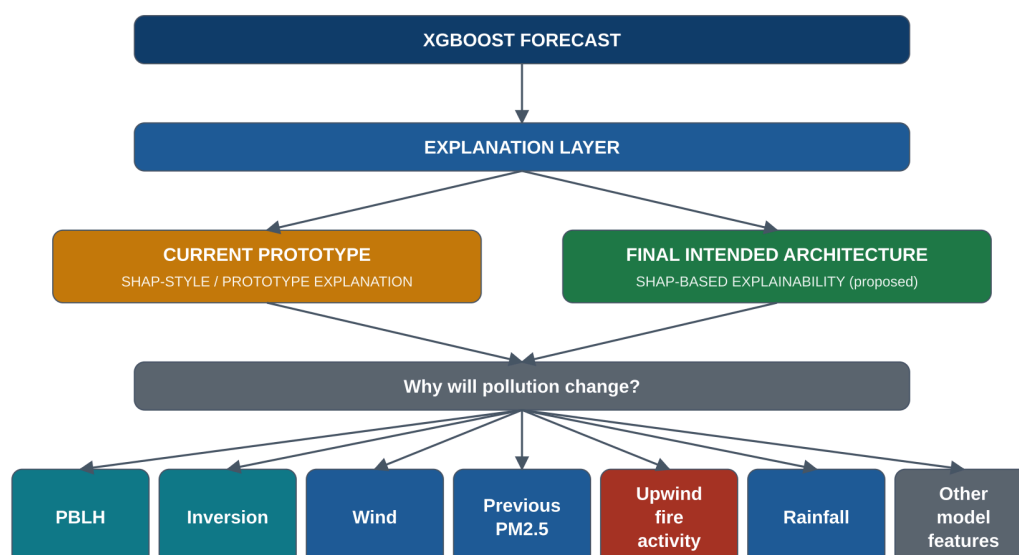


Figure 8: Explainable AI: prototype and intended architecture

Current prototype status. The prototype contains a SHAP-STYLE / PROTOTYPE EXPLANATION layer. It does not yet contain genuine output from the SHAP library, and no SHAP values or plots are presented in this report. SHAP-based explainability is the proposed final architecture.

19. Confidence / Uncertainty

The system includes a confidence-information layer intended to communicate how reliable a forecast may be. Long-horizon forecasts inherently carry more uncertainty, and the interface should make that visible to the user.

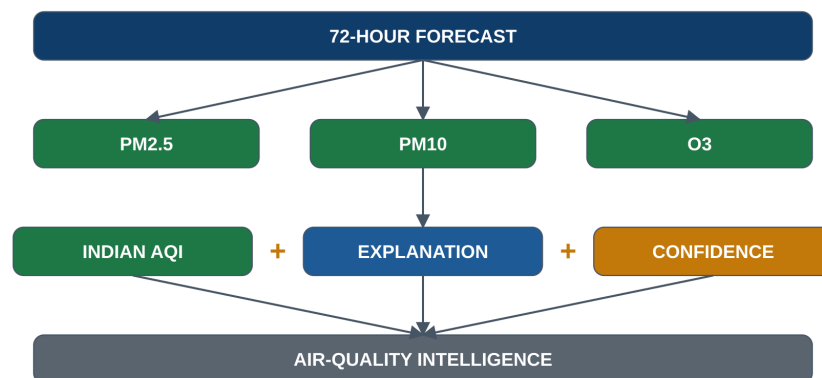


Figure 9: Forecast output, explanation and confidence

Current prototype status. The confidence indicator is a prototype confidence representation. It is not a scientifically validated production uncertainty estimate.

20. Dashboard Design

The proposed dashboard is a decision-support interface built with React, Tailwind CSS, Leaflet/OpenStreetMap and Recharts. Its components are:

- Current AQI and the 24-hour, 48-hour and 72-hour forecasts.
- PM2.5, PM10, O3 and AQI trend charts.
- A Delhi NCR map with air-quality monitoring stations, fire hotspots, wind direction/vectors and pollution zones or heatmap.
- Atmospheric conditions: PBLH, inversion, stability and ventilation/dispersion information.
- Fire transport information, a “Why?” explanation panel and forecast confidence.

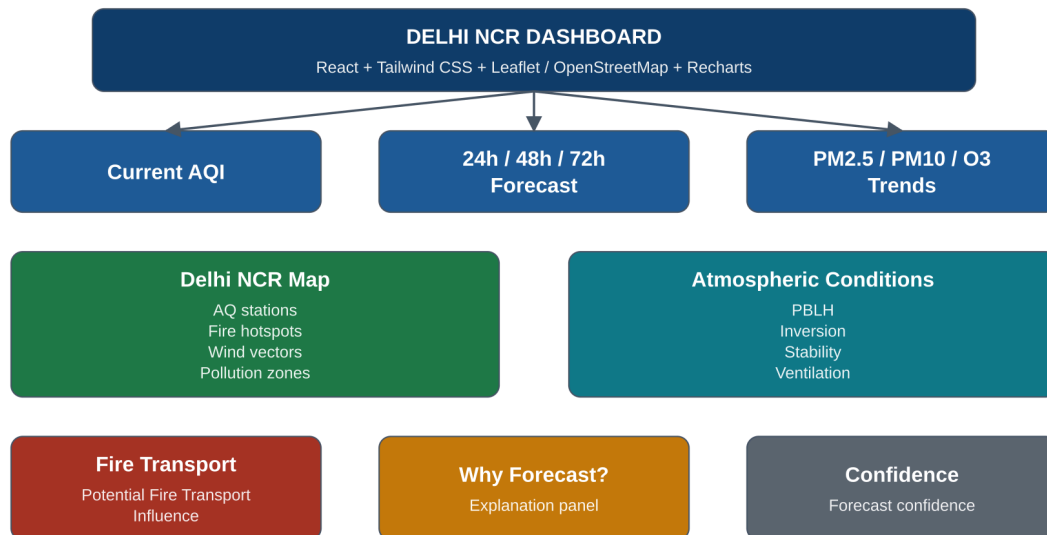


Figure 10: Dashboard architecture

The dashboard supports decisions; it does not guarantee pollution reduction or a particular future AQI.

21. Backend/API Architecture

The backend is written in Python with FastAPI. It is deliberately a single modular application; microservices are not used. Its responsibilities are data processing, feature generation, physics-informed calculations, fire transport calculations, forecast generation, AQI calculation, explanation and confidence information, and dashboard data.

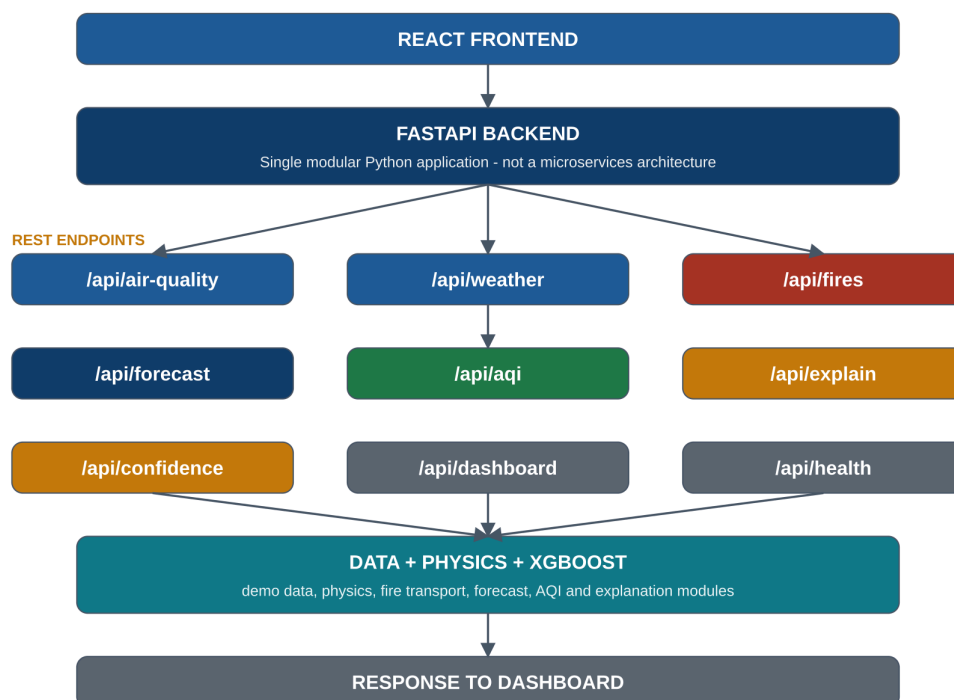


Figure 11: Backend/API architecture

Table 6: Conceptual API endpoints

Endpoint	Purpose
/api/air-quality	Air-quality observations (PM2.5, PM10, O3)
/api/weather	Meteorological information
/api/fires	Fire hotspot information
/api/forecast	72-hour pollutant forecast
/api/aqi	Indian AQI values, category and trend
/api/explain	Explanation information ("Why?")
/api/confidence	Forecast confidence information
/api/dashboard	Combined data for the dashboard
/api/health	Service health check

22. Technology Stack

Table 7: Technology stack

Layer	Technology	Purpose
Frontend	React, Tailwind CSS, Leaflet, OpenStreetMap, Recharts	Interactive dashboard, maps and charts
Backend	Python, FastAPI	Data processing, physics calculations, APIs
Machine learning	XGBoost	Multi-pollutant forecasting
Explainability	SHAP (intended) / SHAP-style prototype explanation	Feature-contribution explanation
Data	CPCB, Open-Meteo, NASA FIRMS, Copernicus ERA5	Air-quality, weather, fire and atmospheric inputs
Prototype	Python, JavaScript, CSV/JSON/demo data	Local demonstration without external databases

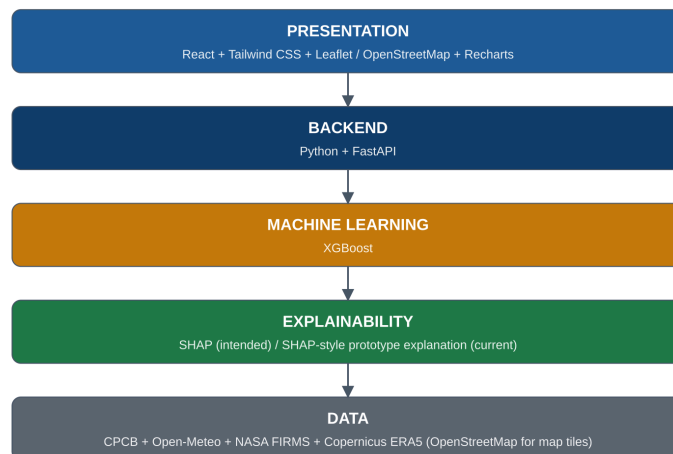


Figure 12: Technology stack

23. Implementation Approach

The implementation is kept simple and modular for a student-level MVP. For the prototype, local CSV/JSON/Python demo data are used; no separate database server is introduced. The current repository structure is:

```
backend/  
  data/demo_data.py  
  services/aqi.py  
  services/explanation.py  
  services/fire_transport.py  
  services/forecast.py  
  services/physics.py  
  main.py  
  requirements.txt  
  test_api.py  
frontend/  
  src/api/client.js  
  src/components/  
  src/pages/  
package.json, package-lock.json, vite.config.js
```

Each backend service corresponds to a functional module: physics (atmospheric indicators), fire_transport (transport index), forecast (XGBoost), aqi (Indian AQI), and explanation (explanation layer). The frontend, built with Vite, reaches the backend through api/client.js.

24. Validation Methodology

Forecasting air quality is a time-series problem, so random splitting is not used as the primary validation approach. The planned validation uses a chronological train/validation/test split, rolling-window validation and expanding-window validation, evaluated separately at horizons of 1h, 6h, 24h, 48h and 72h (12h optionally).

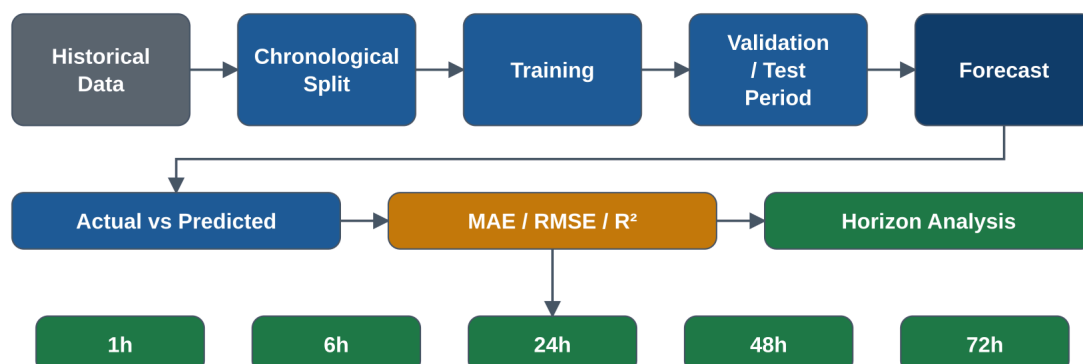


Figure 13: Validation flow

Table 8: Validation metrics

Metric	Purpose	Target
MAE	Average absolute error of PM2.5, PM10, O3 forecasts	To be defined after validation
RMSE	Error measure that penalises large deviations	To be defined after validation
R ²	Proportion of variance explained	To be defined after validation
AQI MAE	Average absolute error of forecast AQI	To be defined after validation
AQI RMSE	Large-error-sensitive AQI error	To be defined after validation
AQI category accuracy	Whether the predicted AQI category matches the observed one	To be defined after validation

Planned visualizations are actual versus predicted PM2.5, actual versus predicted AQI, the 72-hour forecast curve, error versus forecast horizon, feature importance, SHAP visualization (only after genuine SHAP implementation) and a fire-plus-wind transport visualization.

Validation status. Not yet experimentally validated in the current prototype. No metric values, targets or results are claimed in this report.

25. Feasibility and Viability

Technical feasibility rests on open-source technologies (Python, XGBoost, FastAPI, React) and public data. Operationally, the solution is a web-based platform with multi-source integration, interactive visualization and a decision-support interface. For scalability, the modular architecture is designed for Delhi NCR first and can be extended to other Indian cities.

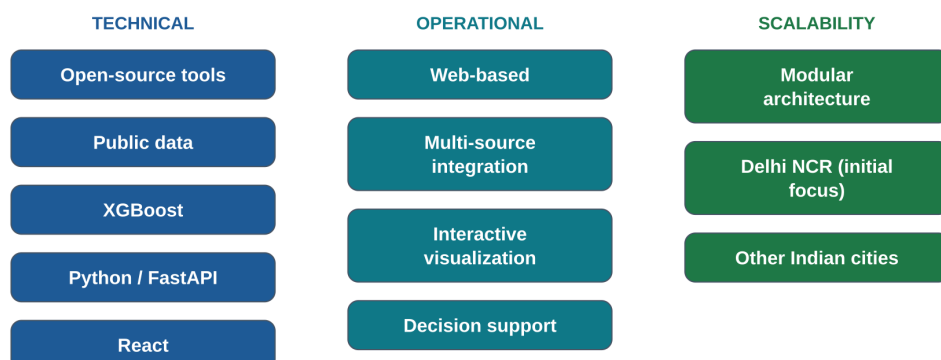


Figure 14: Feasibility framework

26. Challenges and Mitigation

Table 9: Challenges and mitigation

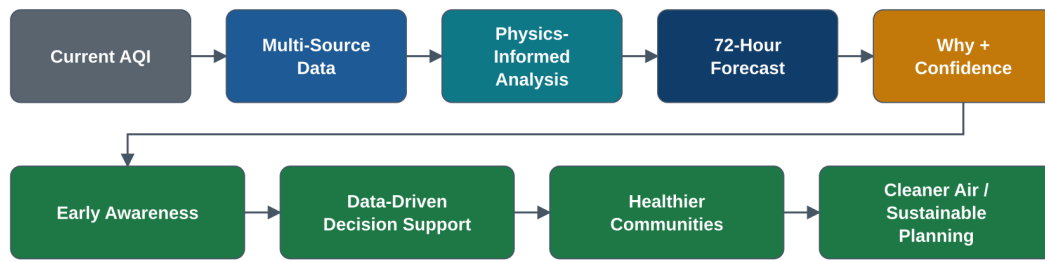
Challenge	Mitigation
Missing or inconsistent data	Data validation and missing-data imputation
Different spatial resolutions	Spatial alignment
Different temporal resolutions	Temporal alignment
API / data-source interruptions	Multiple data sources and fallback sources
Forecast uncertainty	Confidence estimation (validation still required)
Variable fire influence	Wind-based fire transport modelling instead of raw fire counts
Increasing data / user load	Modular / scalable deployment

27. Innovation / USPs

- **Physics-informed AI:** combines atmospheric-science concepts with XGBoost.
- **72-hour forecasting:** moves beyond current-AQI monitoring toward future pollution intelligence.
- **Atmospheric-aware prediction:** uses PBLH, inversion and atmospheric stability.
- **Fire transport intelligence:** uses distance, wind, timing and atmospheric conditions instead of simply counting fires.
- **Pollution–weather coupling:** captures interactions between pollution, weather and atmosphere.
- **Explainable forecasting:** provides information about factors influencing predictions.
- **Confidence-aware forecasting:** communicates forecast confidence as part of decision support.
- **Spatial intelligence:** combines AQ stations, fire hotspots and environmental conditions on a map.
- **Scalability:** adaptable from Delhi NCR to other Indian cities.

28. Impact and Benefits

The system aims to transform today's AQI monitoring into 72-hour air-quality intelligence. Potential benefits include better public awareness, earlier pollution-risk awareness, healthier communities through earlier exposure information, data-driven decision-making for governments and institutions, support for business and industry operational planning, environmental planning, and contribution toward cleaner-air and sustainable-development objectives.



Potential benefits - the system supports decisions; it does not guarantee pollution reduction.

Figure 15: Impact flow

These are potential benefits of better information. The system does not itself guarantee pollution reduction.

29. Current Prototype Status

The backend test currently passes when run with the command below. This confirms the API behaves as tested; it is not a measure of forecasting accuracy.

```
python -m pytest backend/test_api.py -q
```

Table 10 distinguishes the current prototype from the intended final architecture.

Table 10: Current prototype versus intended final architecture

Component	Current prototype	Final / intended architecture
XGBoost	Model instantiated/trained; the displayed forecast currently uses a deterministic prototype timeline	Actual model-based forecast output
Explainability	SHAP-style explanation	Genuine SHAP-based explanation
Confidence	Prototype confidence indicator	Properly validated confidence / uncertainty
Fire transport	Transport / influence index	Improved and validated transport modelling
Data integration	Demo / local data	Integrated public data sources

30. Limitations

- The current forecast output uses a deterministic prototype/demo timeline rather than direct model prediction.
- Genuine SHAP library output is not yet implemented.
- The confidence indicator is not scientifically validated.
- Fire transport represents potential influence or transport risk, not exact plume concentration.
- Real-world performance requires larger historical datasets and proper time-series validation.
- Long-horizon forecasts inherently involve increasing uncertainty.

31. Future Work

- Replace the prototype forecast timeline with actual XGBoost model predictions.
- Implement genuine SHAP library-based explanations.
- Expand historical real-world datasets.
- Perform proper long-horizon validation.
- Improve confidence/uncertainty estimation through validation.
- Improve fire transport modelling with additional validation.
- Extend from Delhi NCR to other Indian cities.

32. Conclusion

This report presented a physics-informed, explainable 72-hour air-quality forecasting and decision-support platform for Delhi NCR. By coupling air-quality observations, meteorology, boundary-layer and inversion conditions and upwind fire transport, and feeding physics-informed features to XGBoost, the design aims to move from measuring today's AQI toward understanding and forecasting tomorrow's air, with an explanation and confidence indication for each forecast.

The current prototype demonstrates the modular backend, the frontend structure and the intended interaction, while its forecast timeline, explanation layer and confidence indicator remain prototype-level. The next steps are model-based forecasting, genuine SHAP explanations, real-data integration and rigorous time-series validation. Until that validation is complete, no performance claims are made.

33. References

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